

Want to have an edge over others?

Master LINUX



Read
LINUX For You

For more info, log on to:

www.linuxforu.com



ASIA'S FIRST
LINUX
MAGAZINE

LINUX
ForYou

Q1. How can I use a supercapacitor in my experimental circuit?

Umesh Purohit

Chanderi, Madhya Pradesh

A1. It is necessary to know the device before you think of connecting it in the circuit. For the benefit of the readers, we describe here the supercapacitor and its working. The references given at the end will give you sufficient reading material to know all about the latest technology of supercapacitors.

A supercapacitor resembles a regular capacitor, except that it offers very high capacitance in a small package. Electric double-layer capacitors, also known as supercapacitors, electrochemical double-layer capacitors or ultracapacitors, are electrochemical capacitors that have an unusually high energy density compared to common capacitors, typically of the order of thousands of times greater than a high-capacity electrolytic capacitor.

For instance, a typical D-cell sized electrolytic capacitor has a capacitance in the range of tens of millifarads. The same-size electric double-layer capacitor would have a capacitance of several farads—an improvement of about two or three orders of magnitude in capacitance, but usually at a lower working voltage.

Larger, commercial electric double-layer capacitors have capacities as high as 5000 farads. The highest energy density in production is 30 Wh/kg. Supercapacitors aim to fill the gap between capacitors and batteries.

In contrast with traditional capacitors, electric double-layer capacitors do not have a conventional dielectric. Rather than two different plates separated by an intervening substance, these capacitors use plates that are, in fact, two layers of the same substrate, and their electrical properties, the so-called 'electrical double layer,' result in the effective separation of charge despite the vanishingly thin physical separation of the layers (of the order of nanometres). The lack of need for

a bulky layer of dielectric permits the packing of plates with much larger surface area into a given size, resulting in their extraordinarily high capacitances in small-size packages.

In an electrical double layer, each layer by itself is quite conductive, but the physics at the interface where the layers are effectively in contact means that no significant current can flow between the layers. However, the double layer can withstand only a low voltage, which means that electric double-layer capacitors rated for higher voltages must be made of matched series-connected individual electric double-layer capacitors—much like series-connected cells in higher-voltage batteries.

Due to the capacitor's high number of charge-discharge cycles (millions or more compared to 200-1000 for most commercially available rechargeable batteries), there are no disposable parts during the whole operating life of the device, which makes the device environment-friendly. Batteries wear out in a few years, and their highly reactive chemical electrolytes present a serious disposal and safety hazard.

Other advantages of electric double-layer capacitors compared with rechargeable batteries are extremely low internal resistance or ESR, high efficiency (up to 98 per cent), high output power, extremely low heating levels and improved safety. According to California-based Institute of Transportation Studies (ITS) test results, the specific power of electric double-layer capacitors can exceed 6 kW/kg at 95 per cent efficiency.

Supercapacitors are being used for transportation applications. China is experimenting with a new form of electric bus (capabus) that runs without powerlines, using power stored in large on-board electric double-layer capacitors, which are quickly recharged whenever the electric bus stops at any bus stop (under so-called electric umbrellas) and fully charged at the terminus.

In 2001 and 2002, VAG, the public transport operator in Nuremberg, Germany, tested a hybrid bus that used a